Horseshoe Forests for High-Dimensional Causal Survival Analysis

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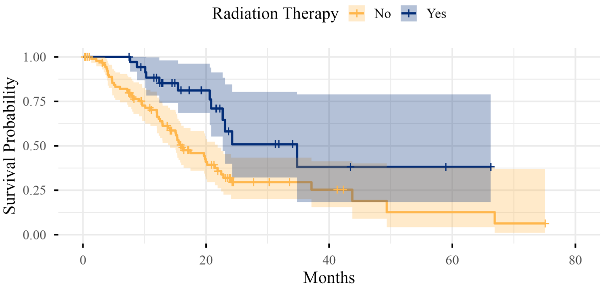
Motivating question: pancreatic cancer

Pancreatic ductal adenocarcinoma (PDAC).

- One of the deadliest cancers; median survival < 6 months.
- Surgical resection + adjuvant radiotherapy is often the only curative option.



TCGA-PAAD cohort.

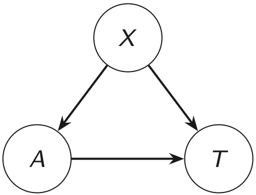
- $n = 130$ resected patients.
- 11 clinical covariates + $\sim$3,000 gene expressions.
- Censoring $\approx 47\%$.

Does adjuvant radiotherapy improve

The setting

Observational survival data $(Y_i, \delta_i, A_i, X_i)$, $i = 1, \ldots, n$:

- $Y_i = \min(T_i, C_i)$, $\delta_i = \mathbf{1}\{T_i \leq C_i\}$ (right-censored).
- $A_i \in \{0, 1\}$ (binary, non-randomised).
- $X_i \in \mathbb{R}^p$ with $p \gg n$.



Three sources of difficulty.

- *Observational*: treatment and outcome share causes.
- *Censored*: only partial information about T.
- *High-dimensional*: classical non-parametric tools break down.

Estimands: potential outcomes

For each individual, imagine *both* treatment counterfactuals:

$$T(1): \text{ survival under treatment.} \qquad T(0): \text{ survival under control.}$$

Fundamental problem of causal inference. Only one is ever observed:

$$T \;=\; A\,T(1) \;+\; (1-A)\,T(0).$$

Estimands on the log scale.

- *Average treatment effect:* $\tau \;:=\; \mathbb{E}[\log T(1) - \log T(0)]$.
- *Conditional average treatment effect:* $\tau(x) \;:=\; \mathbb{E}\big[\log T(1) - \log T(0) \,\big|\, X = x\big]$.

Goal. Estimate $\tau(x)$ — how the treatment effect varies across patients.

Intuition: the acceleration factor

On the log scale, $\tau(x)$ is a difference. Exponentiated, it is a *multiplier on survival time*:

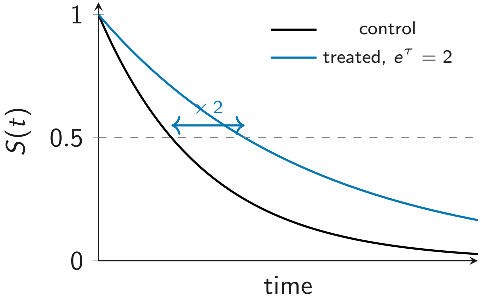
$$T(1) \,\big|\, X = x \;\stackrel{d}{=}\; e^{\tau(x)} \cdot T(0) \,\big|\, X = x.$$

Reading the picture.

- Treatment *stretches the patient's clock* by a factor $e^{\tau(x)}$.
- $e^{\tau(x)} = 2 \;\Rightarrow\;$ median survival doubles.
- Same shape, different time scale.

Contrast with the hazard ratio.

- HR multiplies *hazards*.
- Acceleration factor multiplies *time itself*.



Why AFT, and not Cox?

The default for survival is Cox. Three reasons to switch here:

1. Interpretability. The acceleration factor $e^{\tau(x)}$ acts on the *time scale*, not on hazards. A doubling of expected survival is a directly clinically meaningful statement.

2. Collapsibility.

$$\mathbb{E}_X\big[e^{\tau(X)}\big] \;=\; \text{marginal acceleration factor.}$$

We can meaningfully *average* CATEs to get the ATE. The hazard ratio is *non-collapsible* — marginal HR $\neq$ average of conditional HRs.

3. Robustness. Hougaard (1999): the AFT model is robust to omitted independent covariates; partial-hazard models are not.

Throughout, we model $\log T$.

Unconfoundedness, intuitively

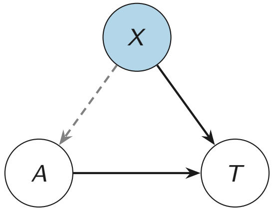
The worry. Sicker patients may be more likely (or less likely) to be treated. A naive treated-versus-untreated comparison conflates the *treatment effect* with *baseline differences*.

Unconfoundedness.

$$T(a) \perp\!\!\!\perp A \,\Big|\, X, \qquad a \in \{0, 1\}.$$

"Within each value of X, treatment looks as if randomised."

Operationally. Imagine matching like-for-like: for any pair of patients with $X_i = X_j$, the only systematic difference between them is the treatment they received.



Conditioning on X (filled node) "cuts" the dashed back-door arrow. Within each X-stratum, the A–T contrast is causal.

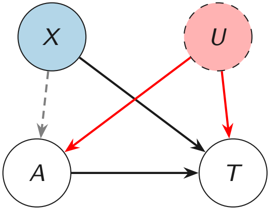
What goes wrong if a confounder is missed?

Now suppose we adjust for X but *miss* an unmeasured confounder U:

- The back-door path $A \leftarrow U \rightarrow T$ remains open.
- Naive comparison still picks up the U effect.
- Estimate is biased — even with infinite data.

Practical consequence in high dimensions.
We dare not drop covariates aggressively: a dropped variable may be a confounder.

This motivates a regularisation strategy that keeps every covariate in the model.



The unmeasured U leaves an open back-door path $A \leftarrow U \rightarrow T$.

The remaining identification assumptions

SUTVA. No interference between units; well-defined potential outcomes. Implies $T = A\,T(1) + (1-A)\,T(0)$.

Positivity. For every x, both treatment arms have positive probability:

$$0 \;<\; e(x) \;:=\; P(A=1 \mid X=x) \;<\; 1.$$

In $p \gg n$ regimes, this can only hold in a lower-dimensional confounding subspace — a working assumption we cannot verify directly.

Independent censoring (familiar):

$$C(a) \;\perp\!\!\!\perp\; T(a) \;\Big|\; X, A.$$

This makes the distribution of $T \mid X, A$ identifiable from (Y, δ).

Under SUTVA + unconfoundedness + positivity + independent censoring, $\tau(x)$ is identified from the observed data.

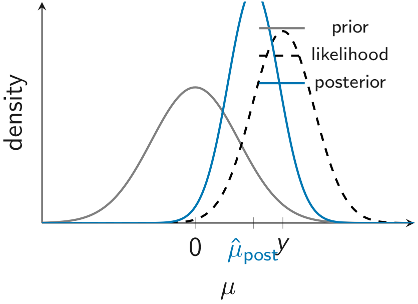
Bayes in one slide: priors regularise

For a single observation $y \sim \mathcal{N}(\mu, \sigma^2)$ with prior $\mu \sim \mathcal{N}(0, \kappa^2)$:

$$\mu \mid y \;\sim\; \mathcal{N}\left(\underbrace{\frac{\kappa^2}{\kappa^2+\sigma^2}}_{\text{shrinkage weight}} \cdot y,\;\; \frac{\kappa^2\sigma^2}{\kappa^2+\sigma^2} \right).$$

Three things to take away.

- The posterior is a *distribution*, not a point estimate.
- The posterior mean is the MLE *pulled toward the prior mean* — the prior regularises.
- A tight prior shrinks more; a vague prior shrinks less.



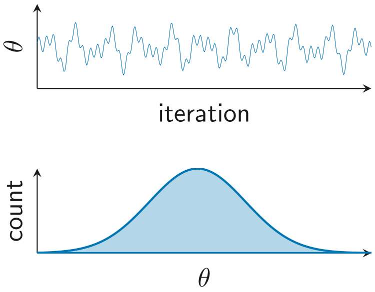
Slogan. A Bayesian model regularises for free, and the posterior carries its own uncertainty.

Posterior inference via MCMC

For models we cannot integrate by hand, we *sample* from the posterior.

Markov chain Monte Carlo (MCMC).

- Build a Markov chain whose stationary distribution is the posterior.
- Run the chain. Discard burn-in.
- Remaining draws are (correlated) samples from the posterior.



Summarising the posterior.

- Posterior mean ≈ average of draws.
- 95% credible interval ≈ 2.5%–97.5% sample quantiles.

"With 95% posterior probability, τ lies in this interval."

The AFT decomposition

We split log event times into a prognostic and a treatment-effect component:

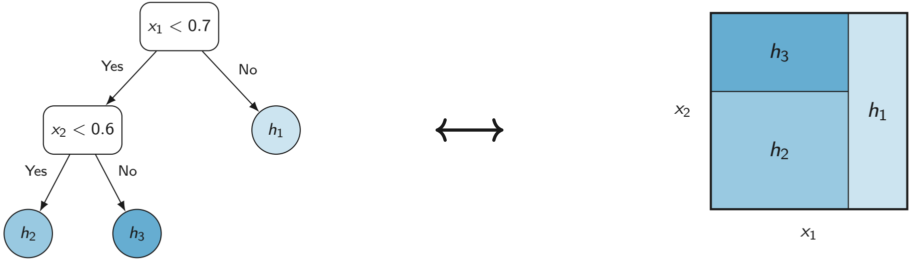
$$\log T(a) \;=\; \underbrace{f\big(x, \hat{e}(x)\big)}_{\text{prognostic}} + \underbrace{a \cdot \tau(x)}_{\text{treatment effect}} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Two regression functions to learn.

- $f(x, \hat{e}(x))$: prognostic baseline; includes the *estimated* propensity score $\hat{e}(x)$ as a balancing covariate (Hahn et al., 2020). Plugging $\hat{e}$ in avoids feedback between the propensity and the outcome models.
- $\tau(x)$: heterogeneous treatment effect — our object of interest.

Plan. Place a flexible Bayesian tree-ensemble prior on each of f and τ, *separately*.

One tree is just a partition



Each tree. A piecewise-constant function on a partition of $\mathbb{R}^p$. Same idea as CART — and as the trees inside a random forest.

BART: a sum of weak learners

Random forest. A handful of *deep* trees fit greedily, then averaged. Variance reduction through bagging.

BART (Bayesian Additive Regression Trees). A *sum* of $m = 200$ *shallow* trees:

$$g(x) = \sum_{j=1}^{m} g(x; \mathcal{T}_j, \mathcal{H}_j).$$

- Each tree is a weak learner — it accounts for only a small piece of the signal.
- Trees correct each other's residuals: closer in spirit to boosting than to bagging.
- Fit by sampling from a posterior, not by greedy splitting.

Where does regularisation come from? The Bayesian prior — explicitly favouring small, shallow trees with small step heights.

The standard BART prior

For each tree, the prior factorises as

$$p(\mathcal{T}_j, \mathcal{H}_j) \;=\; p_{\mathcal{T}}(\mathcal{T}_j)\, p_{\mathcal{H}}(\mathcal{H}_j \mid \mathcal{T}_j).$$

Structure prior $p_{\mathcal{T}}$. A depth-penalised Galton–Watson branching process:

$$P(\text{node at depth } d \text{ splits}) \;=\; a\,(1+d)^{-b}.$$

Discourages deep trees; the splitting variable and split point are chosen uniformly.

Step-height prior $p_{\mathcal{H}}$. Independent Gaussians:

$$h_{j\ell} \;\sim\; \mathcal{N}\big(0, 1/(4\sqrt{m})\big).$$

Variance scaled by $1/\sqrt{m}$ so the sum stays in the observed response range.

All standard-BART regularisation lives in *where the splits go*.

Where could you regularise a tree ensemble?

Two natural places:

1. **In the tree structure $\mathcal{T}_j$.**

   Depth penalty, sparse split rules, Dirichlet on splitting variables (DART, Soft-BART, Shrinkage BCF, . . .).

2. **In the step heights $\mathcal{H}_j$.**

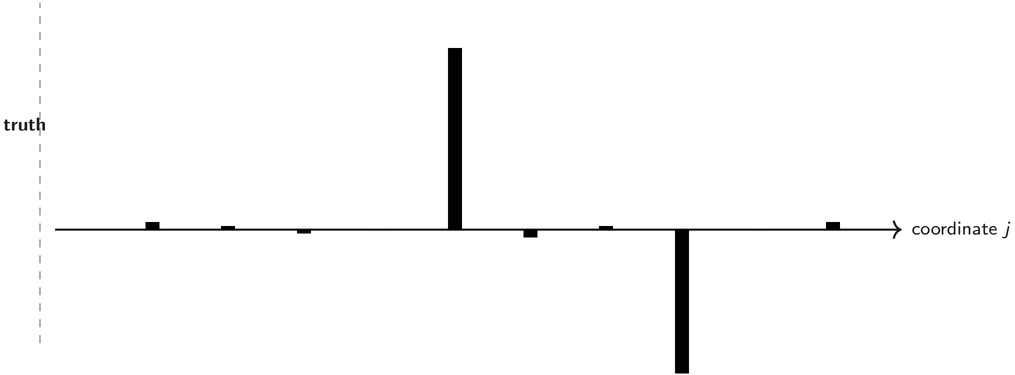
   Shrink the *magnitudes* of the local effects.

Existing high-dimensional work focuses on (1).

Our proposal: regularise through (2), with a horseshoe prior on step heights.

Why shrinkage matters in high dimensions

With $p \gg n$, an unregularised estimator fits noise as eagerly as signal.



- **MLE / no shrinkage:** unbiased but high-variance — noise contaminates every coordinate.
- **Ridge:** stable, but shrinks the large signals *just as much* as the noise.
- **Lasso:** sets small coefficients to exactly zero, but tends to *also* attenuate the large ones.

The horseshoe density

The horseshoe prior on a coefficient θ:

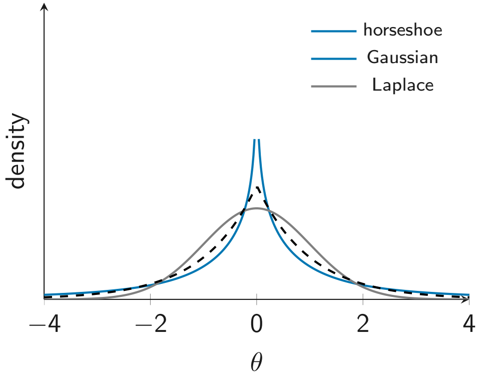
$$\theta \mid \lambda, \gamma \sim \mathcal{N}(0, \lambda^2\gamma^2),$$

$$\lambda, \gamma \sim \mathcal{C}^+(0, 1).$$

Two defining features:

- Pole at zero — strong shrinkage of noise.
- Heavy tails — large signals are *not* shrunk.

Gaussian has neither; Lasso has the spike but not the tails.



Why "horseshoe"? The shape of the shrinkage weight

Reparametrise: define $\kappa = 1/(1+\lambda^2\gamma^2) \in [0,1]$. Then the posterior mean is

$$\widehat{\theta} \;=\; (1-\kappa)\,\bar{y}.$$

$$\kappa = 1 \;\Rightarrow\; \text{full shrinkage to 0;} \qquad \kappa = 0 \;\Rightarrow\; \text{no shrinkage.}$$

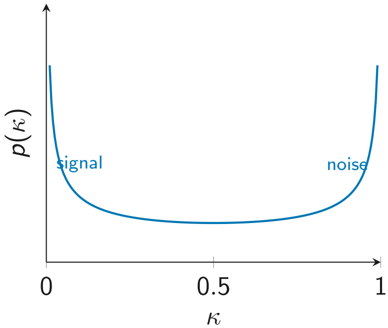
Under the horseshoe prior:

$$\kappa \;\sim\; \mathrm{Beta}\left(\tfrac{1}{2},\tfrac{1}{2}\right).$$

This density is U-shaped. It puts mass at the two extremes:

- $\kappa \approx 1$: "this is noise, kill it."
- $\kappa \approx 0$: "this is signal, let it through."

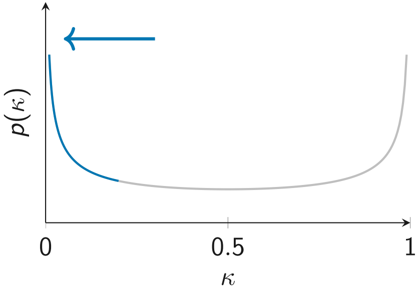
The Bayesian model picks κ *per coefficient*, from the data.



That U-shape is the "horseshoe".

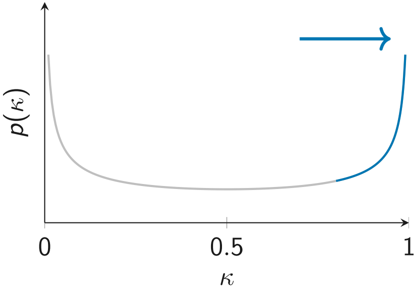
Two regimes, decided coefficient by coefficient

Signal regime: $\kappa \to 0$



$\widehat{\theta} \approx \bar{y}$ — no shrinkage.

Noise regime: $\kappa \to 1$



$\widehat{\theta} \approx 0$ — full shrinkage.

The horseshoe is built to live at the two ends of κ. It does not compromise like Lasso or ridge.

(van der Pas et al., 2014, 2017: near-minimax posterior contraction over sparsity classes.)

Global–local: a global rule with local exceptions

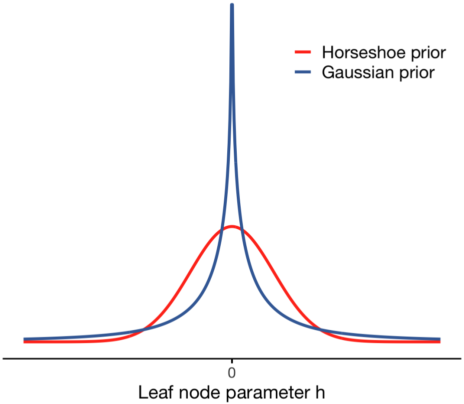
$$h_\ell \mid \lambda_\ell^2, \gamma^2 \sim \mathcal{N}(0, \lambda_\ell^2 \gamma^2), \qquad \lambda_\ell \sim \mathcal{C}^+(0,1), \quad \gamma \sim \mathcal{C}^+(0,1).$$

Two layers of shrinkage.

- **Global γ.** Pulls *everything* towards zero. Learned: if the world is sparse, γ becomes small.
- **Local λ_ℓ.** Lets individual coordinates *escape* the global pull when the data demand it.

"Shrink globally, recover locally."

Why this beats one-size-fits-all. A single ridge λ can either over-shrink the signals or under-shrink the noise. Horseshoe avoids the trade-off.



The novel move: horseshoe on the step heights

For a tree $\mathcal{T}$ with leaves $\ell = 1, \ldots, L$:

$$h_\ell \mid \lambda_\ell^2, \gamma^2 \sim \mathcal{N}(0, \lambda_\ell^2 \gamma^2), \qquad \lambda_\ell \sim \mathcal{C}^+(0,1), \quad \gamma \sim \mathcal{C}^+(0,1).$$

What changes from standard BART.

- Tree structure prior $p_{\mathcal{T}}$: unchanged Galton–Watson.
- Step-height prior: Gaussian $\to$ horseshoe.

Effect. Regularisation migrates from *where splits go* to *how big each local effect is*:

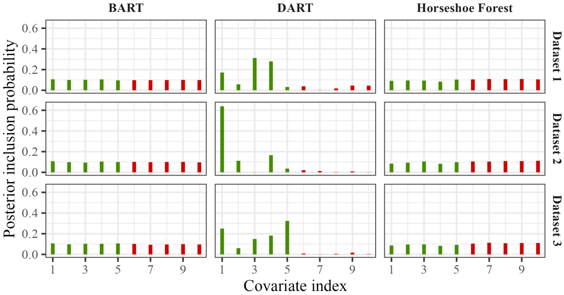
- Trees can split on *any* covariate.
- Step heights are shrunk *per leaf*.
- Real signals: leaves with large step heights. Noise: leaves with step heights near zero.

Why this is better than DART for causal inference

DART (Linero, 2018) regularises the *tree structure*: a Dirichlet prior on splitting probabilities concentrates splits on a subset of covariates.

Two problems.

- DART can concentrate on the *wrong* subset — see green (true signal) being dropped in favour of red (noise) in different replications.
- Dropping a weak confounder violates unconfoundedness ⇒ regularisation-induced confounding (Hahn et al., 2018).



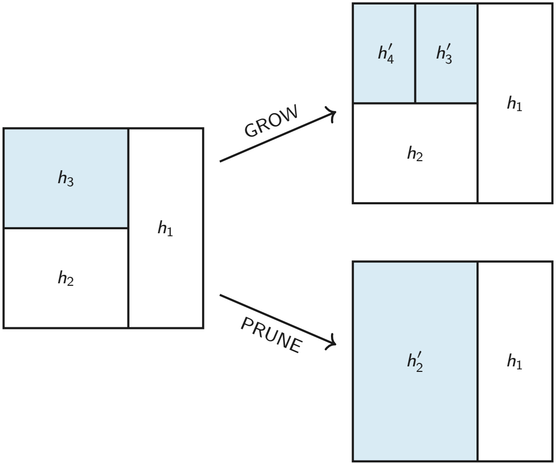
Horseshoe forest. Keeps every covariate split-eligible — shrinks the *magnitudes* of the resulting leaf effects.

Reversible-jump within Gibbs

The horseshoe prior is *not* Gaussian-conjugate $\Rightarrow$ step heights cannot be marginalised out.

GROW and PRUNE moves change the dimension of $\mathcal{H}_j$. We use a reversible-jump step with a tailored proposal for $(\mathcal{T}_j, \mathcal{H}_j)$.

This sits inside a Gibbs sampler that updates the forests f and τ, the error variance σ^2, and the augmented censored event times.



Censoring via data augmentation

We follow Tanner & Wong (1987): treat censored event times as latent variables in the Gibbs sampler.

At iteration t, for each censored i:

$$\log T_i^{(t)} \sim \mathcal{N}\big(\hat{\mu}_i^{(t)}, \sigma^{2,(t)}\big) \quad \text{truncated to } (\log Y_i, \infty),$$

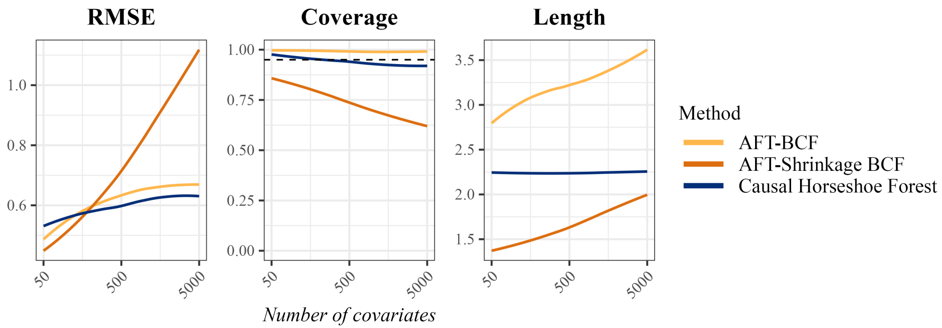
where $\hat{\mu}_i^{(t)} = f^{(t)}\big(x_i, \hat{e}(x_i)\big) + a_i\, \tau^{(t)}(x_i)$.

Net effect. The sampler sees a fully-observed dataset on every iteration. The posterior averages over the uncertainty in the imputed event times.

Propensity score. Estimated separately with a *binary-outcome* Horseshoe Forest (via probit augmentation), then plugged into the prognostic forest. Avoids feedback (Zigler, 2016); improves finite-sample bias (Hahn et al., 2020).

Simulation 1: CATE estimation as p grows

Non-linear treatment effect with sparse main (5%) + interaction (1%) structure; $n = 200$, $p \in [50, 5000]$, censoring $\approx 35\%$.

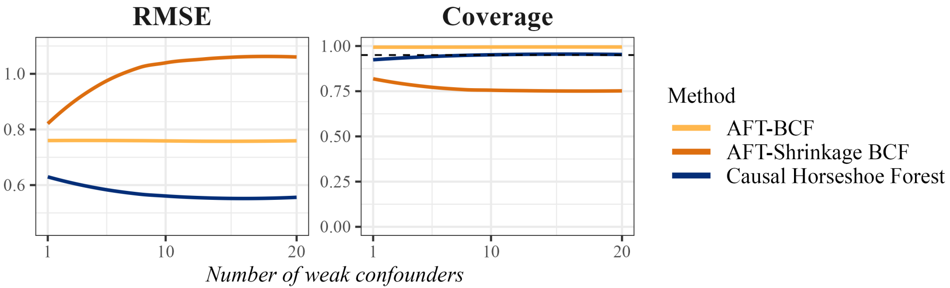


Punchline. Causal Horseshoe Forest: stable RMSE, near-nominal coverage. AFT-BCF: coverage near 1 only because its credible intervals balloon.

Simulation 2: regularisation-induced confounding

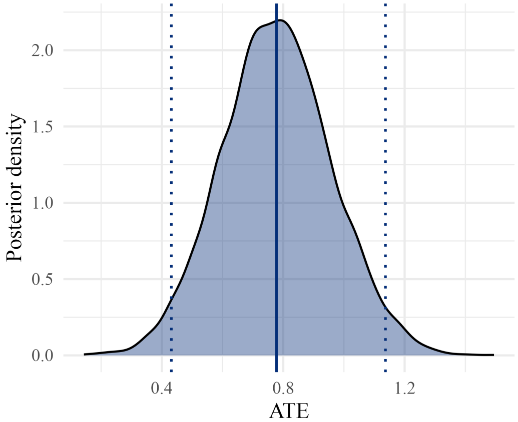
$n = 100$, $p = 500$, censoring $\approx 35\%$. Two groups of confounders:

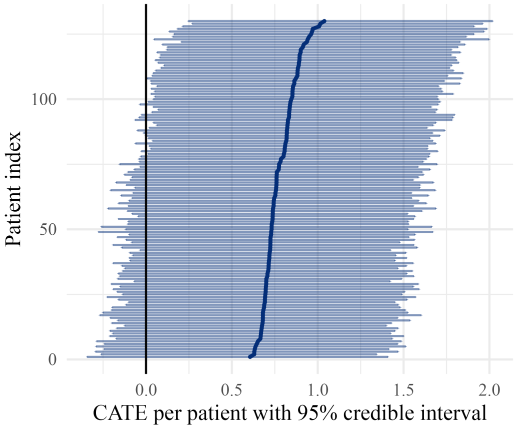
- $|S| = 5$ **strong** confounders — strong on both treatment and outcome.
- $|W| \in \{1, \ldots, 20\}$ **weak** confounders — strong on treatment, weak on outcome.



Punchline. Dirichlet-on-splits (AFT-Shrinkage BCF) drops weak confounders $\Rightarrow$ RMSE blows up. Horseshoe-on-step-heights is stable.

PDAC: results





$\widehat{\text{ATE}} \approx 0.75$, 95% CrI (0.40, 1.11).

Acceleration factor $e^{0.75} \approx 2.1$ — expected survival roughly doubles.

$\approx 96.9\%$ of patients have $P(\tau(X_i) > 0 \mid \text{data}) > 0.95$; little evidence for between-patient heterogeneity.

Caveats. SUTVA not strictly met (regimens vary); positivity unverifiable when $p \gg n$; immortal-time bias may

Take-aways

- Where you regularise matters. Shifting regularisation from the tree *structure* to the *step heights* of a tree ensemble gives a flexible, adaptive prior for high-dimensional CATE estimation.
- The price is a reversible-jump sampler. A tailored tree proposal keeps the chain mixing well.
- Strong empirical performance. Stable RMSE and near-nominal coverage as p grows; robust to regularisation-induced confounding where competing methods break down.
- Available as software. `ShrinkageTrees` on CRAN and GitHub.

Thank you

- Preprint on arXiv: `2507.22004`.
- R package **ShrinkageTrees** on CRAN (and GitHub).
- Questions? `t.jacobs@vu.nl`.

`tijn-jacobs.github.io`

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Funded by the European Union

Simulation 1: data-generating process

$n = 200$, censoring $\approx 35\%$, p swept from 50 to 5000:

$$X_i \sim \mathcal{U}[0,1]^p, \quad A_i \sim \mathrm{Ber}(e(x_i)), \quad \log T_i \sim \mathcal{N}\big(f(x_i) + A_i\tau(x_i),\, \sigma^2\big).$$

$e(x_i) = \Phi(-0.5 + 0.4x_{i1} - 0.1x_{i3} + 0.3x_{i5})$; $f(x_i) = \beta_f^\top x_i$ with β_f spike-and-slab.

Treatment effect. Friedman non-linear core + sparse main effects + sparse pairwise interactions,

$$\tau(x_i) = 10\sin(\pi x_{i1}x_{i2}) + 20(x_{i3} - 0.5)^2 + 10x_{i4} + 5x_{i5} + \beta_\tau^\top x_i + x_i^\top \Gamma x_i,$$

with β_τ spike-and-slab ($s_{\mathrm{main}} = 0.05$) and sparse symmetric Γ ($s_{\mathrm{int}} = 0.01$).

Simulation 2: data-generating process

$p = 500$, $n = 100$, $|S| = 5$ strong confounders, $|W| \in \{1, \dots, 20\}$ weak confounders:

$$\begin{aligned}
e(x_i) &\propto \tfrac{1}{\sqrt{5+|W|}} \left(\textstyle\sum_{j \in S} x_{ij} + 2 \textstyle\sum_{j \in W} x_{ij} \right), \\
f(x_i) &= 4 \sum_{j \in S} x_{ij} + \sum_{j \in W} x_{ij}, \\
\tau(x_i) &= 1 + \sum_{j \in S} \beta_j \, x_{ij}.
\end{aligned}$$

Full conditional update of step heights

Given the current tree $\mathcal{T}$ with L leaves, encode the partition by $D \in \mathbb{R}^{n \times L}$. Then the conditional posterior of $h \in \mathbb{R}^L$ is

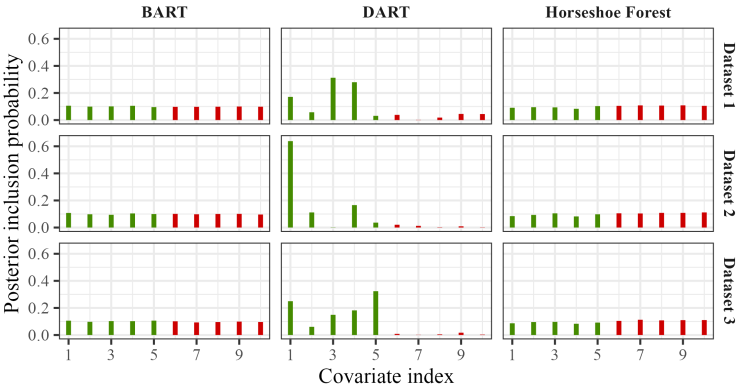
$$h \sim \mathcal{N}_L\Big(\Omega^{-1} D^\top R,\ \Omega^{-1}\Big), \qquad \Omega = D^\top D + \big(\omega\gamma^2 \mathrm{diag}(\lambda_1^2, \ldots, \lambda_L^2)\big)^{-1}.$$

Coordinate-wise:

$$h_\ell \mid \cdot \sim \mathcal{N}\left(\frac{\bar{R}_\ell}{n_\ell + 1/(\gamma^2\lambda_\ell^2\omega)},\ \frac{\sigma^2}{n_\ell + 1/(\gamma^2\lambda_\ell^2\omega)}\right).$$

The shrinkage parameters update in closed form via the inverse-gamma auxiliary representation (Makalic & Schmidt, 2016).

Sparsity on splits can drop real signal



- **BART** & **Horseshoe Forest**: near-uniform splitting mass; horseshoe additionally shrinks noise step heights.
- **DART**: sharp concentration, but across replicates can land on the *wrong* subset.